Register

Register is a group of flip-flops that stores data in binary form.

Based on input/output the register is divided into four categories –

I) Serial input serial output (SISO)

II) Serial input parallel output (SIPO)

III) Parallel input Parallel output (PIPO)

IV) Parallel input serial output (PISO)

Types of register depend on Application –

I) Shift register – SISO, SIPO and PISO.

II) Storage register – PIPO.

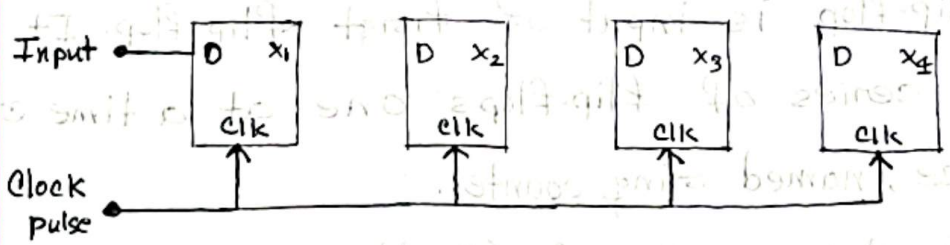
Shift ~~Storage~~ register –

A Shift ~~Store~~ register is a group of flip-flops arranged so that the binary numbers stored in the flip-flops are shifted from one flip-flop to the next for every clock pulse.

Working principle of shift register –

=> A shift register is a sequential circuit made of a series of D flip-flops connected in cascade. Each flip-flop stores one bit of data.

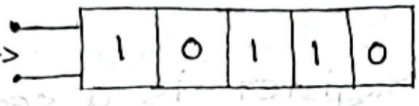
The register operates under the control of a clock signal. With every clock pulse, the data stored in each flip-flop is transfered to the next flip-flop. As a result, the binary data shifts one position from left to right. New ~~The~~ data enters from one end of the register, while the last bit is shifted out. Thus, a shift register stores data and shifts it sequentially with each clock pulse.



Pulse	x_1	x_2	x_3	x_4	
↑	1	–	–	–	
↑	0	1	–	–	
↑	0	0	1	–	
↓	0	0	1	–	(low pulse = 'No shift)
↑	0	0	0	1	

Example -

Data = 10110 ⟶



CLK	X_1	X_2	X_3	X_4	X_5
↑	0	-	-	-	-
↑	1	0	-	-	-
↑	1	1	0	-	-
↓	1	1	0	-	-
↑	0	1	1	0	-
↑	1	0	1	1	0

Ring counter -

Ring counter is typically, a shift register where the output of last flip-flop is input of first flip-flop. It shifts 1 through series of flip-flops one at a time creating a loop. Hence, named ring counter.

—> No. of states = No of flip-flop.

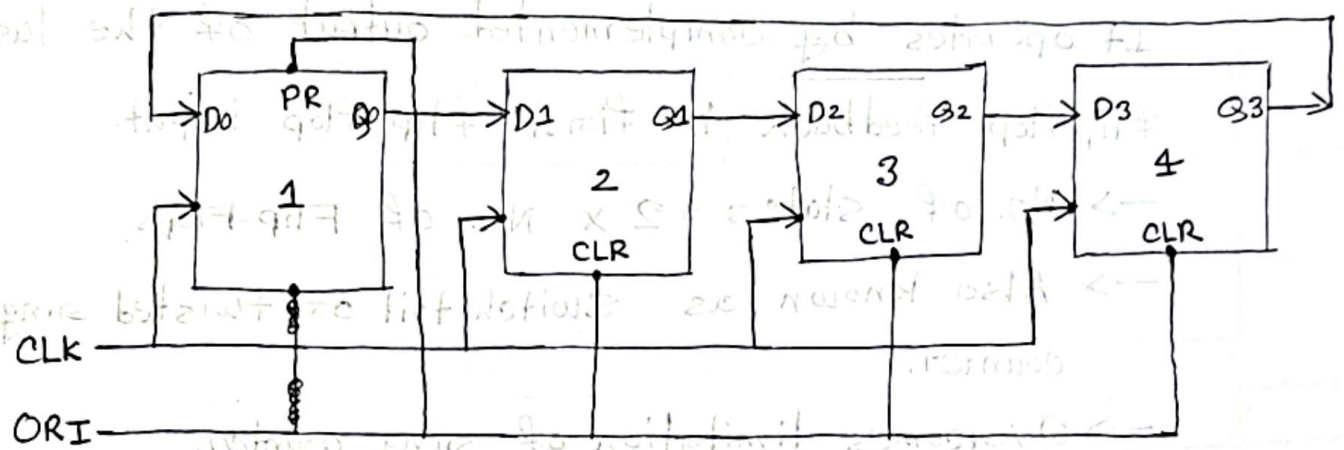
—> Has an active low Over Riding Input (ORI).

—> CLR = Activate → $Q_0 = 1$

—> PR = Activate → $Q_0 = 0$

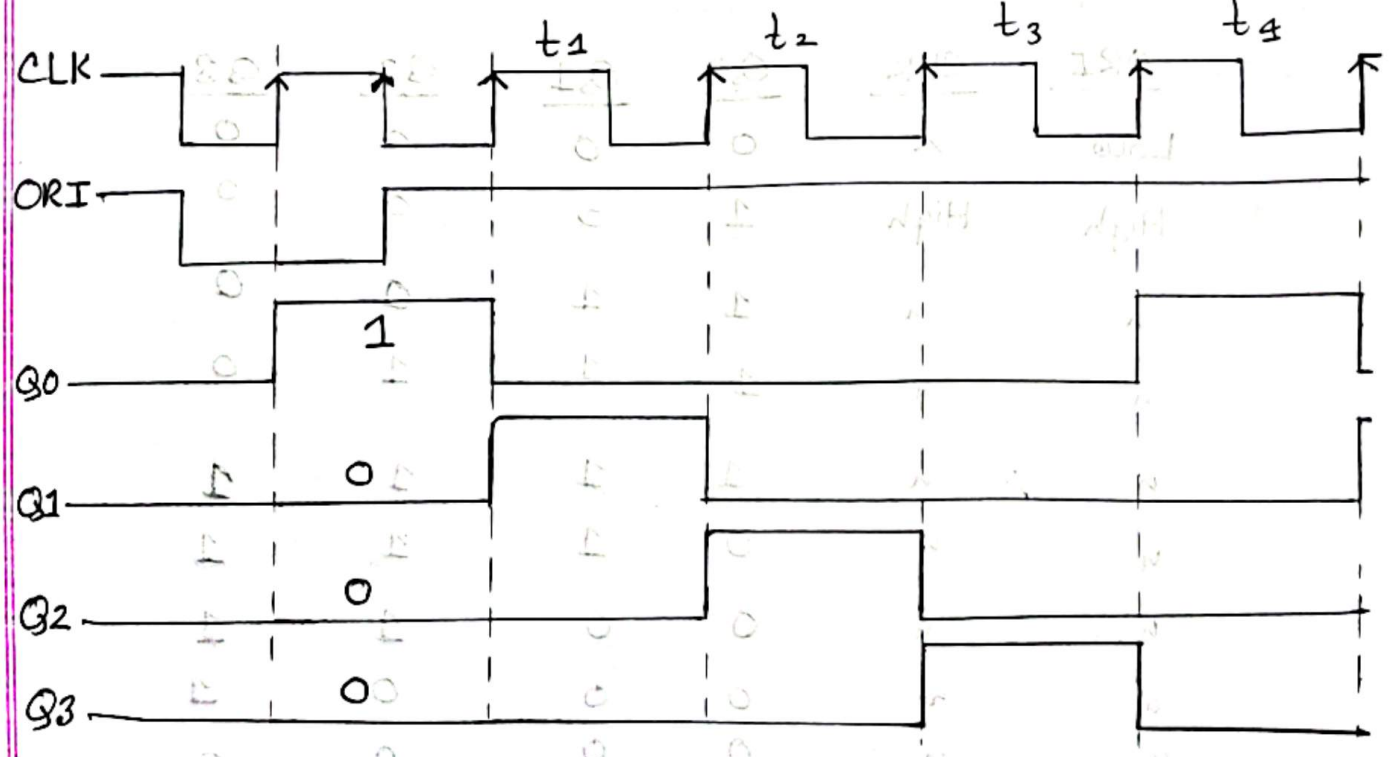
—> Ring counter is used for sequential operation controlling like resistance welding.

4-Bit Ring Counter –



ORI	CLK	Q0	Q1	Q2	Q3
Low	X	1	0	0	0
High	High	0	1	0	0
"	"	0	0	1	0
"	"	0	0	0	1

Graph–



Johnson Counter-

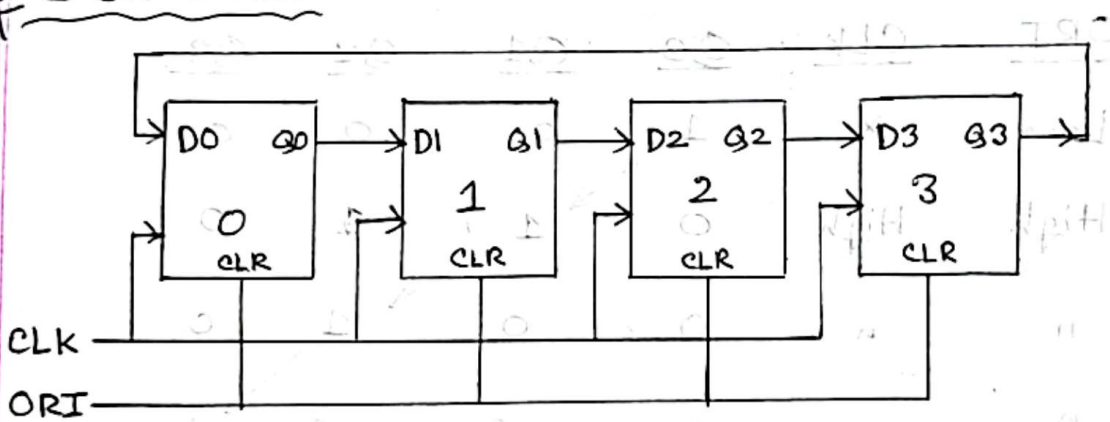
It operates by complemented output of the last flipflop feedback to first flip-flop input.

—> No. of states = 2 x No. of Flip-Flops.

—> Also known as Switch tail or twisted ring counter.

—> Overcomes limitation of ring counter.

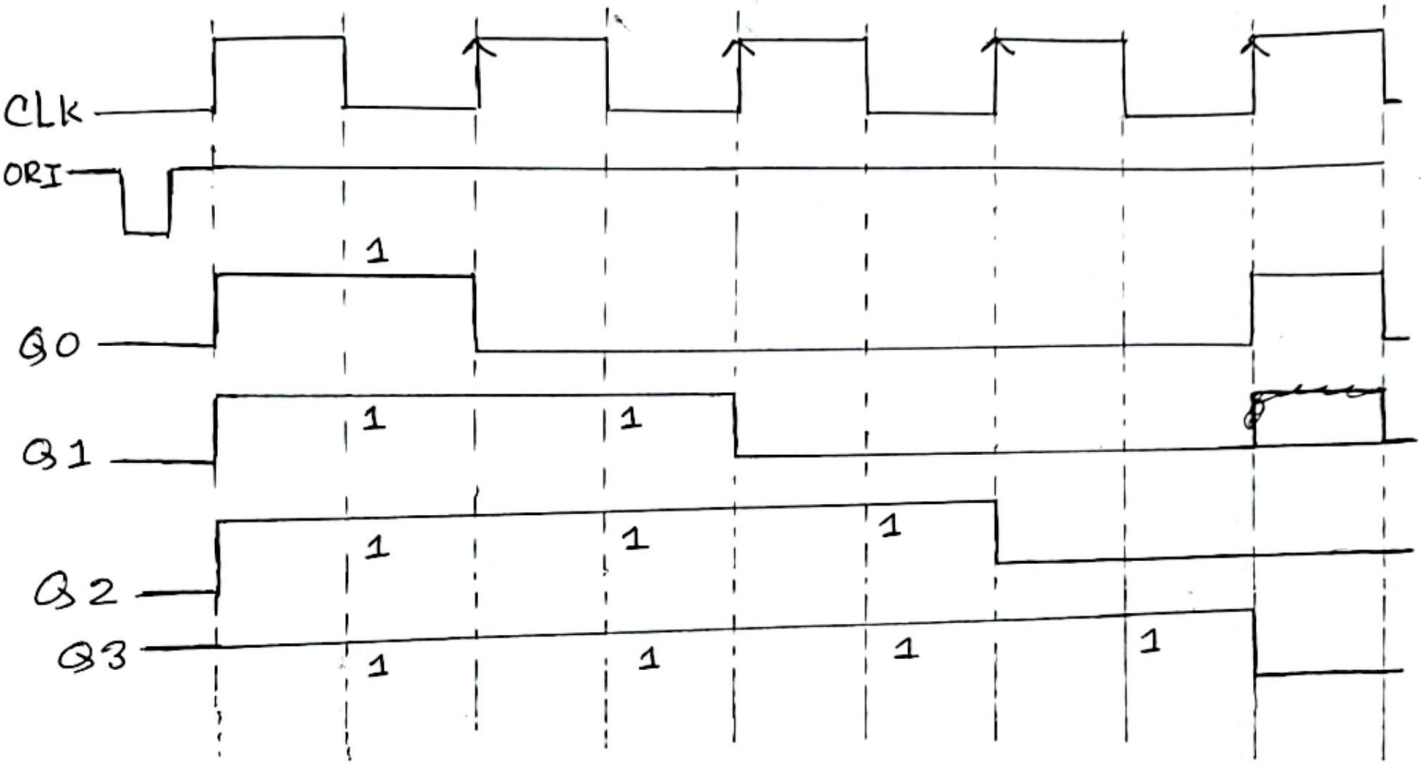
4 Bit Counter-



Truth Table -

ORI	CLK	Q0	Q1	Q2	Q3
Low	X	0	0	0	0
High	High	1	0	0	0
"	"	1	1	0	0
"	"	1	1	1	0
"	"	1	1	1	1
"	"	0	1	1	1
"	"	0	0	1	1
"	"	0	0	0	1
"	"	0	0	0	0

Graph-



Ring Counter vs Johnson Counter-

Ring Counter	Johnson Counter
1. Output of last flip-flop is input in first flip-flop.	1. Complement of last flip-flop is input in first flip-flop.
2. No. of states = No. of flip-flops.	2. No. of states = 2 x NO. of flip-flops.
3. Only 1 bit can be high at a time.	3. The output can be of any pattern.
4. It counts binary sequence.	4. It can not count binary sequence.

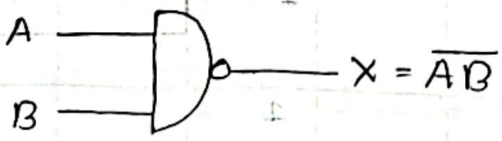
Universal Gate

Universal Gate

1) NAND Gate

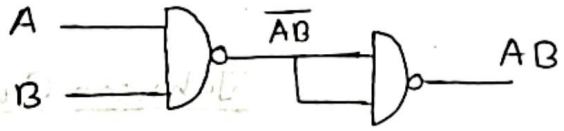
11) NOR Gate

NAND Gate –

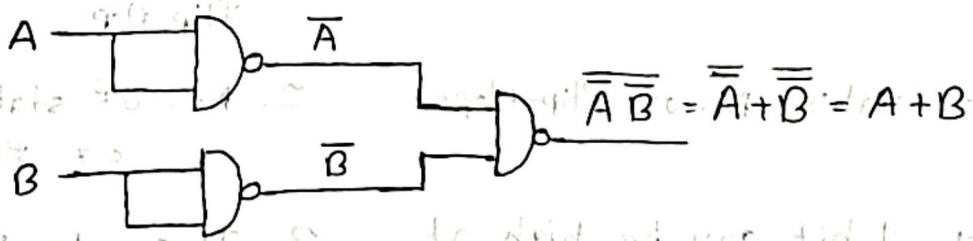


A	B	AB	$\overline{AB}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

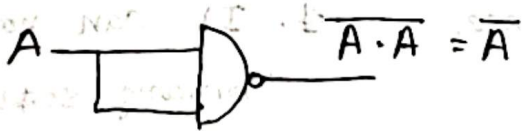
AND Gate –



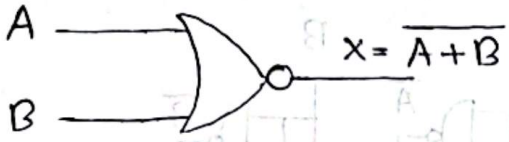
OR Gate –



NOT Gate –

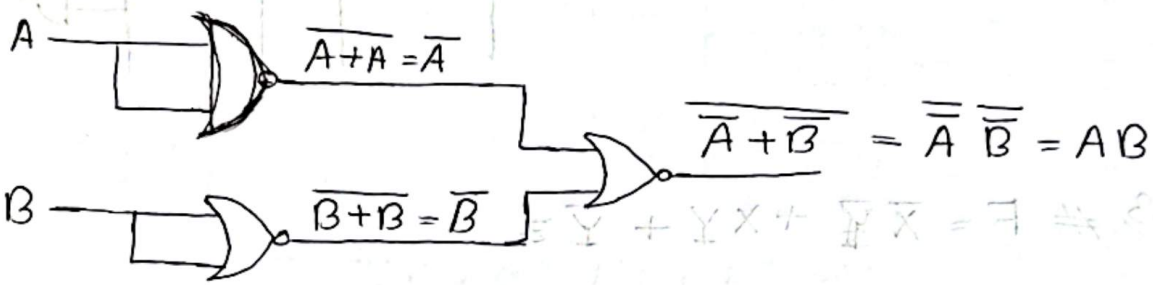


NOR Gate –

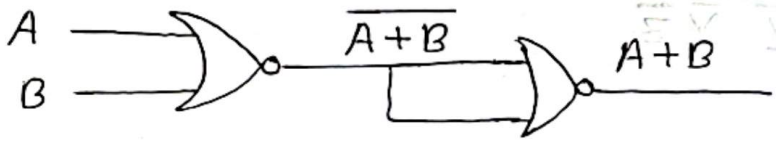


A	B	A+B	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

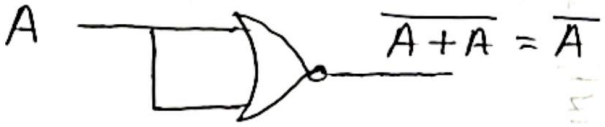
AND Gate –



OR Gate –



NOT Gate –

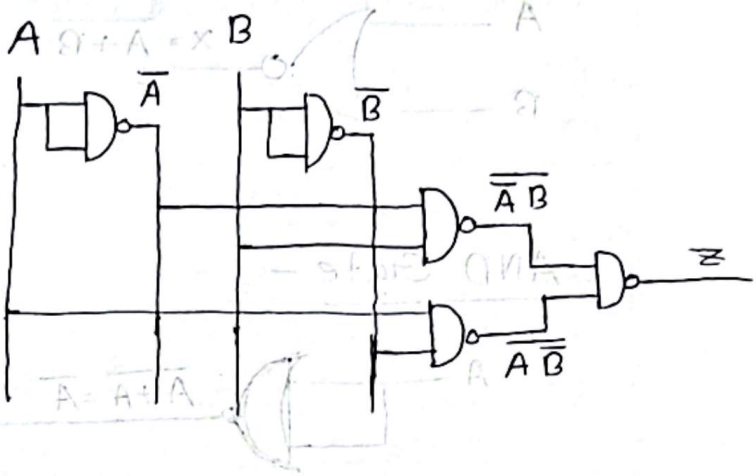


Implement using NAND Gate – $A'B + AB'$

$$Z = A'B + AB'$$

$$= \overline{\overline{A'B + AB'}}$$

$$= \overline{\overline{(A'B)}\ \overline{(AB')}}$$



Prob-3

$F = \overline{X}\overline{Y} + XY + \overline{Y}Z$

$$= \overline{\overline{\overline{X}\overline{Y} + XY + \overline{Y}Z}} \longrightarrow \text{Using NAND}$$

$$= \overline{\overline{\overline{X}\overline{Y}}\ \overline{XY}\ \overline{\overline{Y}Z}}$$

$F = \overline{X}\overline{Y} + XY + Y\overline{Z}$

$$= \overline{\overline{\overline{X}\overline{Y} + XY + Y\overline{Z}}}$$

$$= \overline{\overline{\overline{X}\overline{Y}}\ \overline{XY}\ \overline{Y\overline{Z}}}$$

$$= \overline{(X+Y)\ (\overline{X}+\overline{Y})\ (Y+\overline{Z})} \longrightarrow \text{Using NOR}$$

$$= \overline{(X+Y)} + \overline{(\overline{X}+\overline{Y})} + \overline{(Y+\overline{Z})}$$

$$= \overline{\overline{(X+Y)} + \overline{(\overline{X}+\overline{Y})} + \overline{(Y+\overline{Z})}}$$

Prob 1

Implement using NOR Gate, $F = \overline{W}\,\overline{x} + \overline{x}\,\overline{z}$

=>

$$F = \overline{W}\,\overline{x} + \overline{x}\,\overline{z}$$

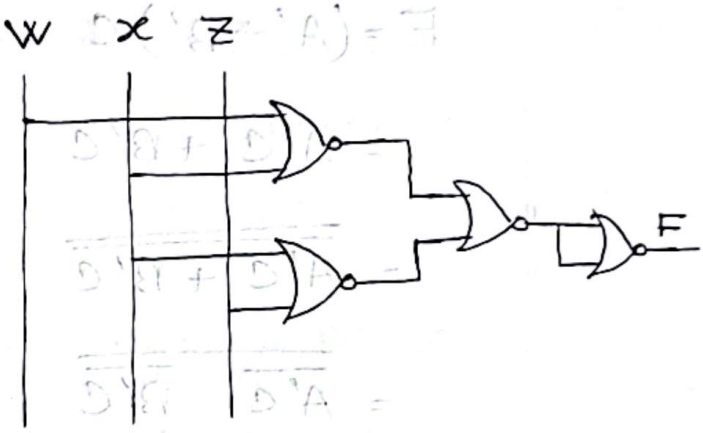
$$= \overline{\overline{\overline{W}\,\overline{x} + \overline{x}\,\overline{z}}}$$

$$= \overline{(\overline{\overline{W}\,\overline{x}})\,(\overline{\overline{x}\,\overline{z}})}$$

$$= \overline{(W+x)\,(x+z)}$$

$$= \overline{(W+x)} + \overline{(x+z)}$$

$$= \overline{\overline{(W+x)} + \overline{(x+z)}}$$



Prob 2

Implement using NAND Gate, $F = A(AB' + C)$

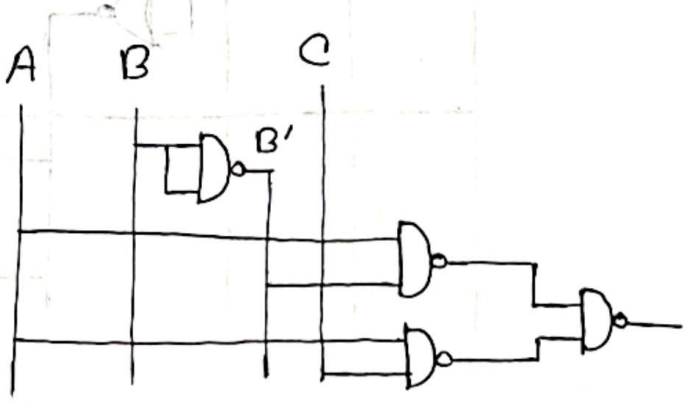
=>

$$F = A(AB' + C)$$

$$= AB' + AC$$

$$= \overline{\overline{AB' + AC}}$$

$$= \overline{\overline{AB'}\;\overline{AC}}$$



Prob 4

$F = (A' + B')C$, implement using NOR Gate.

=>

$$F = (A' + B')C$$

$$= A'C + B'C$$

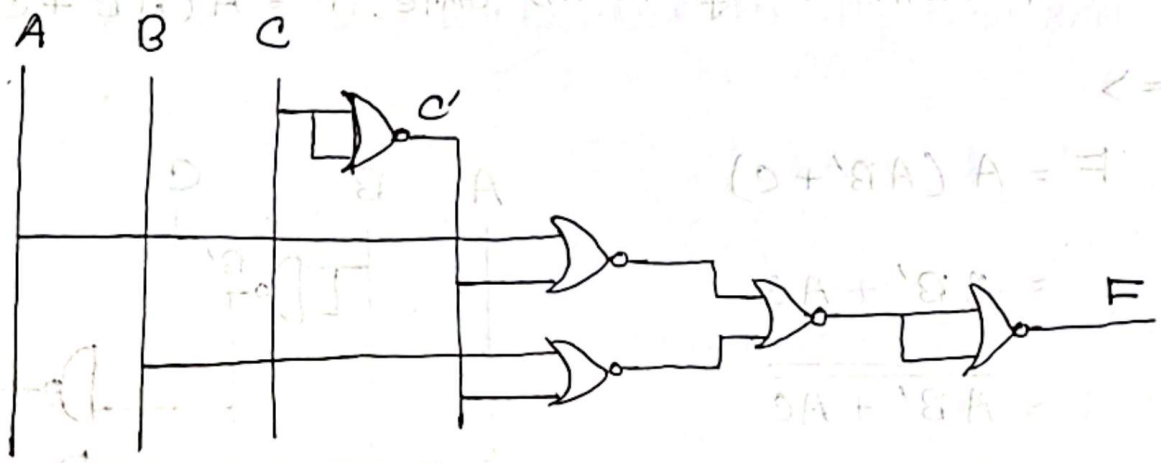
$$= \overline{\overline{A'C + B'C}}$$

$$= \overline{\overline{A'C}\ \ \overline{B'C}}$$

$$= \overline{(A + C')(B + C')}$$

$$= \overline{(A + C')} + \overline{(B + C')}$$

$$= \overline{\overline{(A + C')} + \overline{(B + C')}}$$



Memory Devices

Memory is the electronic component in a computer or electric device that acts as a temporary holding place for data.

—> Main memory :-

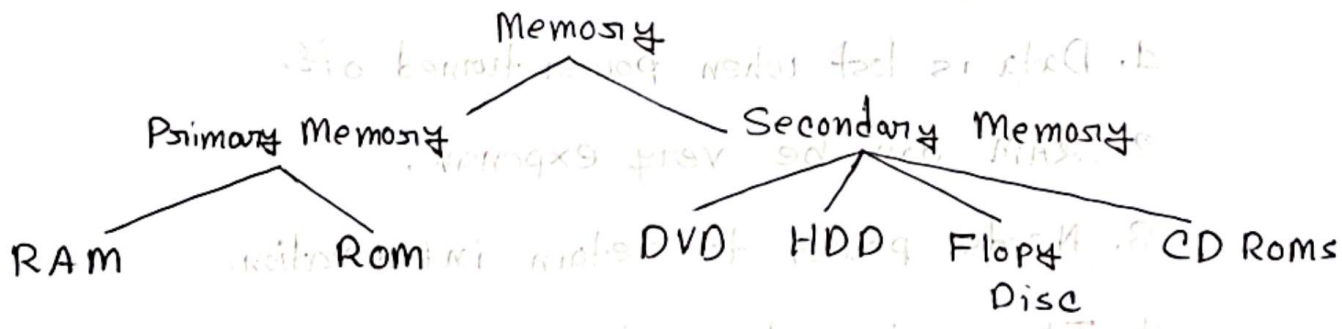
Semiconductor memories are used as main memory devices. Also called working memory or primary memory. Constant communicate with CPU.

Ex- RAM, ROM

—> Auxilary memory :-

Also known as secondary memory or storage memory. Stores data that is currently not being used by the CPU. Stores massive amount of data but slower than main memory.

Ex- SSD, HDD, CD etc.



RAM (Random Access Memory) –

It is a volatile memory which stores data temporarily. If power turned off, the data is erased. Stores the data that is being used during the execution of a program.

——> High speed performance
——> Volatile nature
——> Multitasking
——> System program & instruction execution

Advantages –

1. RAM is faster than any other type of storage.
2. RAM allows a computer to handle multiple applications simultaneously.
3. CPU can read & write data directly from RAM during program execution.

Disadvantages –

1. Data is lost when power turned off.
2. RAM can be very expensive.
3. Needs power to retain information.
4. It requires large physical space.

Two types of RAM-

1. Static RAM (SRAM) -
   Uses flip-flops circuits to store data.
   Retains information as long as power is supplied without refresh.

2. Dynamic RAM (DRAM) -
   Uses capacitors to store data.
   Must be periodically refreshed to maintain information.

Difference -

SRAM	DRAM
1. Uses flip-flops to store data.	1. Uses capacitors to store data.
2. Consumes more power than DRAM.	2. Power efficient than SRAM.
3. Used in cache memory for quick acess.	3. Used in main memory for large data storage.
4. Its faster.	4. Its slower.
5. More expensive	5. Less expensive.

ROM (Read Only Memory) –

Its a non-volatile memory. which stores data permanently Doesn't depends on power on/off. Has decoder and OR gates that process data and store in registers.

Advantages :-

1. Retains data even when power is off.
2. Data remains secure.
3. Data doesn't accidently get erased.
4. Low production cost.
5. Quick data access.

Disadvantages :-

1. Data can't be easily modified.
2. Slow writing compared to RAM.
3. Not suitable for large amount of data storage.

Types of ROM-

1. Masked ROM (MROM) -

It's information stored at the time the integrated circuit is manifactured.

Made of rectangular blocks of transistors.

Can't be reprogrammed by the user.

2. Programmable ROM (PROM) -

Manufactured blanked and can be programmed by the user.

Can be programmed only once, can't be erased or reprogrammed.

Has fusible-link PROM structure, fuses stores data as 0/1.

3. Erasable Programmable ROM (EPROM) -

Can be programmed by the user.

Can be erased and reprogrammed.

Storage element is a mos transistor and can be erased using ultraviolet light.

4. Electrically Erasable Programmable ROM (EEPROM) -

Can be electrically erased.

Reprogrammed as many times.

More compact & improved performance than EPROM.

MOS transistor stores data using high voltage.

RAM vs ROM -

RAM	ROM
1. Data can be easily modified.	1. Data can be modified.
2. RAM is faster than Rom.	2. Rom is slower than RAM.
3. RAM is volatile.	3. Rom is non-volatile.
4. RAM is expensive.	4. ROM is comparatively cheaper.
5. Used for storeing data currently processed by CPU	5. Used to store firmware and BIOS data.
6. Enables read & write operation	6. Allows only read operations.

Other memory device -

1. Cache memory -

High speed semiconductor memory placed between CPU & main memory.

Stores copies of frequently accessed instruction and data so the processor can retrive them with very low latency.

Very small size & fast processing, volatile like SRAM.

2. HDD (Hard Disk Drive) -

Secondary memory used for storing large information.

Magnetic secondary storage device.

Non volatile, stores backups, archives and OS programs.

3. SSD (Solid State Drive) -

A flash-memory based secondary storage device.

Stores data electromically.

SSD faster than HDD.

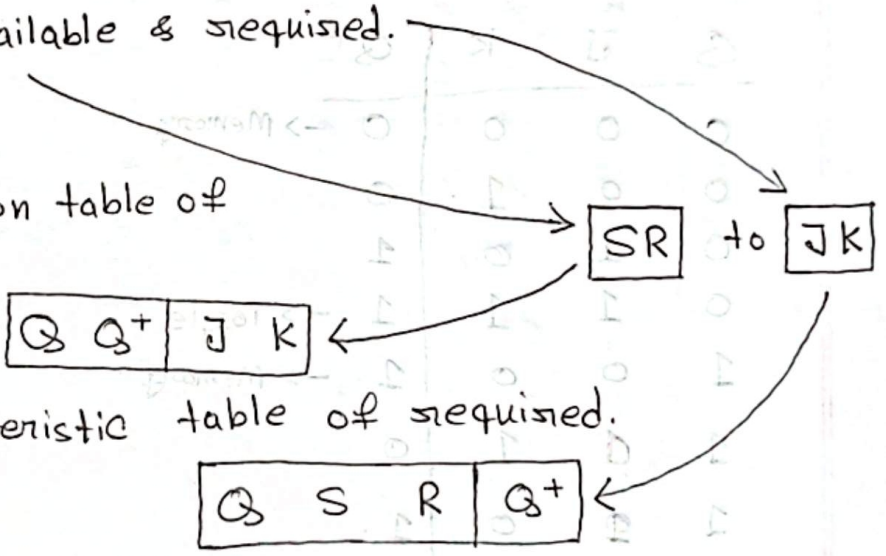
Data is stored as electrical charge in memory cells.

More silent, durable, expensive than HDD.

Flip-Flop Conversion

Steps for conversion-

1. Identify available & required.
2. Make excitation table of available.
3. Make characteristic table of required.
4. Karnaugh-Map Solve.
5. Derive the Boolean expression.
6. Draw circuit.



—o—

SR characteristic & excitation table –

Characteristic Table –

Q	S	R	Q+	
0	0	0	0	→ Memory
0	0	1	0	
0	1	0	1	
0	1	1	x	→ Invalid
1	0	0	1	→ Memory
1	0	1	0	
1	1	0	1	
1	1	1	x	→ Invalid

Excitation Table:-

Q	Q+	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

JK characteristic & excitation table -

Characteristic table-

Q	J	K	Q^+	
0	0	0	0	-> Memory
0	0	1	0	
0	1	0	1	
0	1	1	1	-> Toggle
1	0	0	1	-> Memory
1	0	1	0	
1	1	0	1	
1	1	1	0	-> Toggle

Excitation table-

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

D characteristic & excitation table-

Characteristic table -

Q	D	Q^+	
0	0	0	-> Memory
0	1	1	
1	0	0	
1	1	1	

As input, same output

Excitation table -

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

T Characteristic table & excitation table-

Characteristic table -

Q	T	Q^+	
0	0	0	-> Memory
0	1	1	-> Toggle
1	0	1	-> memory
1	1	0	-> Toggle

Excitation table -

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

SR to JK -

Available = SR ; Required = JK

SR Characteristic table-

Q	S	R	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

SR excitation table-

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

JK characteristic table-

Q	J	K	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

New Merge table –

Q	J	K	Q^+	S	R
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	1	X	0
1	0	1	0	0	1
1	1	0	1	X	0
1	1	1	0	0	1

For S –

Q \ JK	00	01	11	10
0	.		1	1
1	X			X

Q	J K
0	1 1
0	1 0
$\bar{Q}$	J

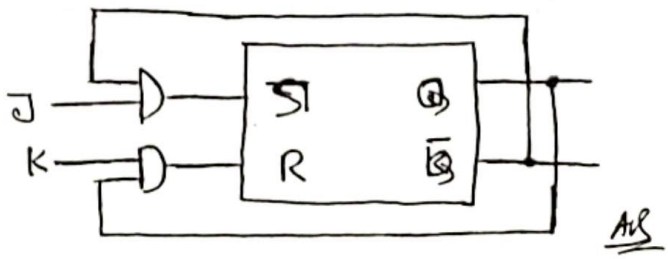
$\therefore S = \bar{Q}J$

For R –

Q \ JK	00	01	11	10
0	X	X		
1		1	1	

Q	J K
1	0 1
1	1 1
Q	K

$\therefore R = Q K$



Ans

JK to SR -

Available - JK ; Required = SR

JK characteristic table-

Q	J	K	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

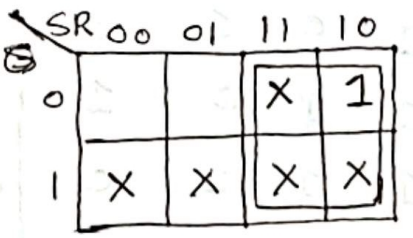
JK excitation table-

Q	Q^+	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

SR characteristic table- (Merge)

Q	S	R	Q^+	J	K
0	0	0	0	0	x
0	0	1	0	0	x
0	1	0	1	1	x
0	1	1	x	x	x
1	0	0	1	x	0
1	0	1	0	x	1
1	1	0	1	x	0
1	1	1	x	x	x

For J,

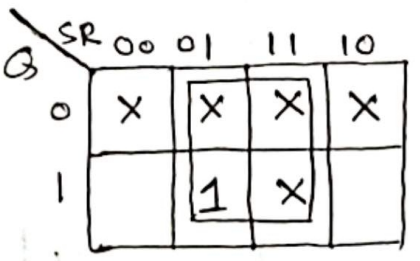


→

Q	S	R
0	1	1
0	1	0
1	1	1
1	1	0

$\therefore J = S$

For k,

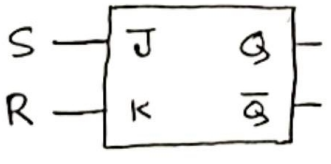


→

Q	S	R
0	0	1
0	1	1
1	0	1
1	1	1

R

$\therefore K = R$



JK to D –

Available = JK ; Required = D

JK excitation table –

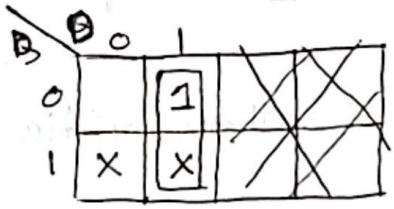
Q	Q^+	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

D characteristic table –

Q	D	Q^+	J	K
0	0	0	0	x
0	1	1	1	x
1	0	0	x	1
1	1	1	x	0

So,

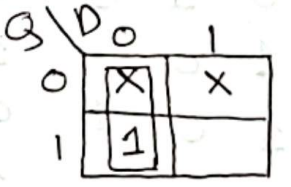
For J,



Q \ D	0	1
0		1
1	x	x
	D	

$\therefore J = D$

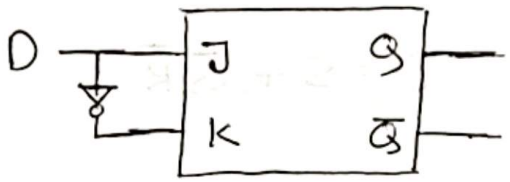
For K,



Q \ D	0	1
0	x	x
1	1	
	$\bar{D}$	

$K = \bar{D}$

$\therefore$ Circuit,



Ans

D to SR -

Available - D ; Required = SR

D excitation table -

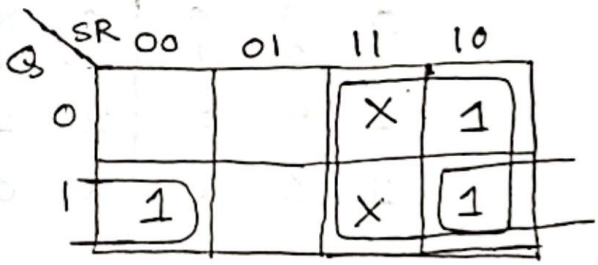
Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

SR characteristic table -

Q	S	R	Q^+	D
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1 ✓
0	1	1	X	X ✓
1	0	0	1	1 ✓
1	0	1	0	0
1	1	0	1	1 ✓
1	1	1	X	X

So,

For D,



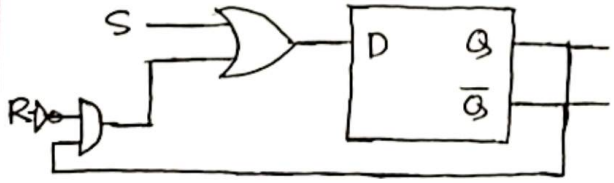
Q S R
0 1 1
0 1 0
1 1 1
1 1 0

S

Q S R
1 0 0
1 1 0

Q $\bar{R}$

$$D = S + Q\bar{R}$$



D to T-

Available = D ; Requisred = T

D excitation table-

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

T Characteristic table-

Q	T	Q^+	D
0	0	0	0
0	1	1	1 ✓
1	0	1	1 ✓
1	1	0	0

So,

For D

Q \ T	0	1
0		1
1	1	

$\therefore D = \bar{Q}T + Q\bar{T}$

$= Q \oplus T$

Circuit-

